

PRODUCTION OF MATCHOUT-DEUTERATED CHOLESTEROL AND ITS DERIVATIVES FOR NEUTRON SCATTERING STUDIES

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ABSTRACT

Selective deuteration of biomolecules is a powerful strategy for neutron scattering studies, enabling contrast variation experiments that reveal the structure and dynamics of complex biological systems. In this work, we present a robust protocol for the production of matchout-deuterated cholesterol, possessing a neutron scattering length density matched to 100% D₂O, together with deuterated sterol intermediates from the cholesterol biosynthesis pathway. The sterols, including cholesterol, 7-dehydrocholesterol (7-DHC), zymosterol, cholesta-5,7,14,24(25)-tetraenol, and cholesta-7,24(25)-dienol, were produced in a genetically engineered *Pichia pastoris* strain grown in deuterated media with non-deuterated glycerol as the carbon source to precisely control the level of deuteration.

Mass spectrometry and ¹H NMR confirmed approximately 89% deuteration of cholesterol, corresponding to 41 deuterium and five protium atoms. The neutron match point was experimentally verified using small-angle neutron scattering. The resulting matchout cholesterol was applied in neutron reflectometry studies of supported phosphatidylcholine lipid bilayers comprising saturated and unsaturated lipids with varying acyl chain lengths. These measurements demonstrated that cholesterol adopts an asymmetric distribution within the membrane, residing closer to the phospholipid headgroup region than to the bilayer center, irrespective of phosphatidylcholine composition. Preliminary infrared and Raman spectroscopy further indicate that individual sterol intermediates produce distinct effects on membrane packing when incorporated into lipid bilayers.

The availability of both hydrogenated and selectively deuterated cholesterol and sterol biosynthetic intermediates significantly expands the experimental toolbox for neutron scattering. These reagents enable contrast-matched investigations of membrane organization, sterol biosynthesis, lipid exchange, lipoprotein interactions, and membrane protein structure and function, particularly when combined with other matchout-deuterated membrane components. More broadly, this work demonstrates a simple and effective approach to controlling biomolecular deuteration through carbon source selection, providing versatile probes for advanced structural studies of biologically relevant membrane systems.

REFERENCES: 1) Waldie, S., Moulin, M., Porcar, L. et al. The Production of Matchout-Deuterated Cholesterol and the Study of Bilayer-Cholesterol Interactions. *Sci Rep* 9, 5118 (2019).